DIVIDED POWER DIFFERENTIALS VIA THE KERNEL OF THE COPRODUCT

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file crystalline.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original lemma 07HH: sources/stacks-chow/tags.txt, source lines 9552–9552. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(A, I, \gamma) \to (B, J, \delta)$ be a homomorphism of divided power rings. Let $(B(1), J(1), \delta(1))$ be the coproduct of (B, J, δ) with itself over (A, I, γ), i.e., such that*

$$\begin{array}{ccc} (B, J, \delta) & \longrightarrow & (B(1), J(1), \delta(1)) \\ \uparrow & & \uparrow \\ (A, I, \gamma) & \longrightarrow & (B, J, \delta) \end{array}$$

is cocartesian. Denote $K = \mathrm{Ker}(B(1) \to B)$. Then $K \cap J(1) \subset J(1)$ is preserved by the divided power structure and

$$\Omega_{B/A,\delta} = K/\left(K^2 + (K \cap J(1))^{[2]}\right)$$

canonically.

Proof. The fact that $K \cap J(1) \subset J(1)$ is preserved by the divided power structure follows from the fact that $B(1) \to B$ is a homomorphism of divided power rings.

Recall that K/K^2 has a canonical B-module structure. Denote $s_0, s_1 : B \to B(1)$ the two coprojections and consider the map $\mathrm{d} : B \to K/(K^2 + (K \cap J(1))^{[2]})$ given by $b \mapsto s_1(b) - s_0(b)$. It is clear that d is additive, annihilates A, and satisfies the Leibniz rule. We claim that d is a divided power A-derivation. Let $x \in J$. Set $y = s_1(x)$ and $z = s_0(x)$. We will use δ instead of $\delta(1)$ for the divided power structure on $J(1)$. We have to show that $\delta_n(y) - \delta_n(z) = \delta_{n-1}(y)(y - z)$ modulo $K^2 + (K \cap J(1))^{[2]}$ for $n \geq 1$. The equality holds for $n = 1$. Assume $n > 1$. Note that $\delta_i(y - z)$ lies in $(K \cap J(1))^{[2]}$ for $i > 1$. Calculating modulo $K^2 + (K \cap J(1))^{[2]}$ we have

$$\delta_n(z) = \delta_n(z - y + y) = \sum\nolimits_{i=0}^{n} \delta_i(z - y)\delta_{n-i}(y) = \delta_{n-1}(y)\delta_1(z - y) + \delta_n(y)$$

This proves the desired equality.

Let M be a B-module. Let $\theta : B \to M$ be a divided power A-derivation. Set $D = B \oplus M$ where M is an ideal of square zero. Define a divided power structure

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on $J \oplus M \subset D$ by setting $\delta_n(x+m) = \delta_n(x) + \delta_{n-1}(x)m$ for $n > 1$, see Lemma 07HH. There are two divided power algebra homomorphisms $B \to D$: the first is given by the inclusion and the second by the map $b \mapsto b + \theta(b)$. Hence we get a canonical homomorphism $B(1) \to D$ of divided power algebras over (A, I, γ). This induces a map $K \to M$ which annihilates K^2 (as M is an ideal of square zero) and $(K \cap J(1))^{[2]}$ as $M^{[2]} = 0$. The composition $B \to K/K^2 + (K \cap J(1))^{[2]} \to M$ equals θ by construction. It follows that d is a universal divided power A-derivation and we win. $\square$